SpatialNet: Wearer-Aware, On-Device Localisation of Pointed-At Objects for Smart Glasses

ANONYMOUS AUTHOR(S)

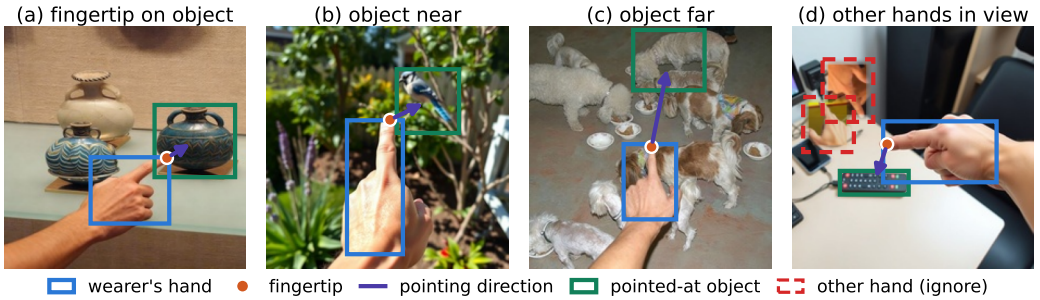


Fig. 1. From one frame of the glasses' camera, SpatialNet finds the wearer's hand, the fingertip, the pointing direction and the pointed-at object, for objects touched (a), near (b) and far (c), and ignores other people's hands (d; the hands in (d) are synthetic, pasted with the procedure of Section 4.6). EgoPoint test images. [The overlays currently show ground-truth labels; replace them with SpatialNet predictions from `predictions/test.csv` after the main run.]

Pointing is the most direct way to ask smart glasses about something in view ("what is *that*?"). Before any assistant can answer, the glasses must work out which object the wearer is pointing at, from their own camera, quickly, and without being misled by other people's hands. The most accurate pointing-grounding models are transformers and multimodal LLMs with hundreds of millions to billions of parameters, built for GPUs, and some receive the pointing hand as an input. We present SpatialNet, a 0.72M-parameter network (0.23 GFLOPs, 1.6 MB in FP16) that, in one forward pass over a single frame, finds the wearer's hand, the fingertip, the pointing direction and the box of the pointed-at object. Its heads form an instance-anchored chain: each stage reads a shared feature map at the location the previous stage predicted, and the target search is conditioned on ray coordinates computed from the model's own fingertip and direction. To prefer the wearer's hand without labelled bystander hands, we train with pasted, unlabelled distractor hands and an ownership-weighted heatmap penalty. On the EgoPoint test split, SpatialNet localises the pointed-at object with IoU ≥ 0.5 in xx.x% of frames, against 93.9% reported for a model with over 300 times as many parameters, and runs in xx.x ms per frame on a smart-glasses developer unit. [Adjust the last sentence to the final T1/T2 results; keep it to one sentence.]

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **Gestural input**; • **Computing methodologies** → *Object detection.*

Additional Key Words and Phrases: smart glasses, pointing gesture, deictic reference, egocentric vision, on-device inference, hand ownership

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM 2474-9567/2026/10-ART
https://doi.org/10.1145/nnnnnnn.nnnnnnn

ACM Reference Format:
Anonymous Author(s). 2026. SpatialNet: Wearer-Aware, On-Device Localisation of Pointed-At Objects for Smart Glasses. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 1, 1 (October 2026), 22 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Camera-equipped smart glasses are becoming everyday devices, and an assistant on them is most useful when the wearer can refer to things in view. Pointing is the most direct way to do so: "what is *that*?", "remind me to buy *this*". Systems have combined speech with pointing since Bolt's *Put-that-there* [4], and recent wearable assistants use pointing and gaze to resolve such pronouns [19]. Before any of this works on glasses, the device must answer a spatial question from its own camera: *which object is the wearer pointing at?*

Answering this on glasses is hard for three reasons.

- *The egocentric view.* The camera sees the wearer's hand from behind and often only in part. The target may be touched or across the room, and a pointing direction drawn in the image is only a rough guide to it.
- *Other people's hands.* A companion's hand or a stranger's pointing gesture can be in view, and must not be mistaken for the wearer's.
- *The budget.* Glasses-class processors share a battery of a few hundred milliampere-hours with the camera, audio and radio. Sending frames to a cloud model adds network latency and moves imagery of bystanders off the device.

Prior work. Pointing-target grounding in egocentric images is an active research area, and EgoPoint-Ground [22] provides a large benchmark for it. The strongest methods are large models:

- multimodal LLMs that reason step by step from the hand to a ray and then to the target [22, 23];
- a 226M-parameter transformer that models the pointing ray explicitly [21].

Some benchmark tasks even give the model the ground-truth hand box [22]. These systems reach high accuracy on a GPU, but none of them is designed for glasses-class hardware, and to our knowledge none addresses other people's hands in view. Egocentric fingertip systems, including one that runs on a phone, stop at the fingertip [16, 17] and never localise the target.

SpatialNet. We present SpatialNet, a 0.72M-parameter convolutional network (0.23 GFLOPs, 1.6 MB in FP16). From a single frame and in one forward pass, it predicts the wearer's hand, the fingertip, the pointing direction and the box of the pointed-at object, together with presence gates for each. Three ideas make this work at such a small size:

- *Instance-anchored heads.* The heads form a chain in which each one reads a shared feature map at the location the previous one predicted, so the fingertip, direction and target all belong to the same hand.
- *Ray-conditioned target search.* The target heatmap is conditioned on six ray-coordinate maps, computed from the model's own fingertip and direction, through a zero-initialised branch. The ray acts as a learned prior that the network uses only as far as it helps.
- *Ownership by training.* Training pastes unlabelled hands of other people into the frames and penalises the hand heatmap for responding to them. The model thus learns to follow the wearer's hand without any labelled bystander hands.

Evaluation. We train and test on EgoPoint and report:

- accuracy by fingertip–target distance, which matters because about 70% of targets are touched by the fingertip;
- comparisons with zero-shot vision–language models on our test images, with published state-of-the-art results, and with simple fingertip-based references;
- ablations of each component;
- ownership on images with pasted and with real hands of other people;
- latency on a smart-glasses developer unit.

[One sentence with the headline T1/T2 numbers, phrased as an accuracy–cost trade-off. For example: "SpatialNet reaches xx.x% IoU≥0.5, against 93.9% reported for VistaRef with over 300× more parameters, and runs in xx.x ms on the glasses."]

Contributions.

(1) *A pointed-target localiser small enough for smart glasses.* It works from the image alone, in one forward pass, through instance-anchored heads with a peak-windowed soft-argmax and a zero-initialised, ray-conditioned target prior (Section 4).
(2) *A training recipe for wearer-awareness without bystander labels.* It combines distractor-hand pasting that follows egocentric geometry, self-pasting, and ownership-weighted heatmap negatives (Section 4.6).
(3) *An evaluation that exposes where pointing localisation fails.* It stratifies results by distance, measures ownership on synthetic and real frames, and places accuracy next to on-glasses cost (Sections 5–7). [Add the user study if T6 is done.]

2 Related Work

Pointing as input. Deictic gestures have been part of multimodal interfaces since *Put-that-there* [4]. Studies of mid-air pointing show that both the hand and the eye matter. Ray-casting models of pointing have systematic offsets [29], and taking the dominant eye and handedness into account makes them markedly more accurate [32]. On wearables, GazePointAR combines gaze, pointing and conversation history to resolve spoken pronouns in AR [19]. EgoPointVQA asks questions about pointed-at objects in egocentric video and feeds 3D hand keypoints, estimated by an off-the-shelf model, to a multimodal LLM as hand-intent tokens [7]. These systems show the value of pointing on wearables; SpatialNet supplies the localisation step that such systems need, at a cost that fits on the glasses.

Fingertips and pointing directions. EgoFinger detects the hand and fingertip in egocentric video for air-writing [16]. GestARLite runs a hand detector, a fingertip regressor and a gesture classifier on a smartphone [17]. DeePoint recognises pointing and estimates its 3D direction from third-person video of the whole body [30]. None of these localises the object being pointed at.

Pointing-target grounding.

- *Third-person.* YouRefIt grounds targets from language and gesture together [6].
- *EgoPoint-Ground.* For egocentric images, it introduced 15K images (real captures from smart glasses, plus synthetically composed and generated images) and several tasks [22]. In its pointing-only task, methods receive the image and the ground-truth hand box. Its SV-CoT baseline reasons with a multimodal LLM from the pointing direction to a ray and then to the target.
- *PointVG-R.* It builds a similar hand–direction–ray–target chain of thought into Qwen2.5-VL-7B with reinforcement learning [23], and names latency as a limitation for wearables.
- *VistaRef.* It adds hand-keypoint and geometric-ray modules and an orientation-consistency loss to a BEiT-3 grounding model (226M parameters with the Base backbone) [21]. With

Table 1. Methods that localise pointing or its target. *Hand box*: the ground-truth hand box is an input. Parameter counts as reported; ">1B": built on a multimodal LLM.

Method	View	Test-time input	Output	Params	On device	Other hands
EgoFinger [16]	ego	image	fingertip	–	–	–
GestARLite [17]	ego	frames	fingertip, gesture	16.3 MB	phone	–
DeePoint [30]	3rd	video	pointing, 3D direction	–	–	–
YouRefIt [6]	3rd	image + language	target box	–	–	–
SV-CoT (POG) [22]	ego	image + hand box	target box	>1B	–	–
PointVG-R [23]	ego	image + instruction	target box	7B	–	–
VistaRef [21]	ego	image + fixed prompt	target box, keypoints	≥226M	–	–
SpatialNet	ego	image	hand, tip, ray, target	0.72M	glasses	trained

the Large backbone, it reports IoU>0.5 for 93.9% of the 3,072 EgoPoint test images; our test split omits 43 of these (Section 5).

- *EgoPoint-Bench.* It shows that general multimodal LLMs still struggle with egocentric pointing [20].

Explicit ray reasoning is thus not new. What SpatialNet changes is the operating point:

- the ray enters a 0.72M-parameter CNN as a learned log-prior on a heatmap, inside one forward pass;
- the model needs no hand box and no prompt;
- it is trained to ignore other people's hands.

Table 1 summarises the differences.

Hands of the wearer and of others. Egocentric hand datasets have long distinguished the camera wearer's hands from other people's: EgoHands labels *my* and *your* left and right hands [2]. Large-scale hand detectors model hand side and contact state [33]. Pointing datasets, EgoPoint included, contain only the wearer's hand. We therefore synthesise other people's hands with copy-paste augmentation [9, 13]. Unlike in standard copy-paste, the pasted instances are deliberately left unlabelled and penalised.

Efficient localisation and grounding. Our trunk follows efficient detection design [28, 36]. Our heads predict keypoint heatmaps in the manner of CenterNet [43], with a differentiable soft-argmax [35]. Open-vocabulary detectors and multimodal LLMs can ground text queries [1, 26, 31, 40], but pointing is not text, and their size rules them out for continuous use on glasses.

3 Problem Definition

Outputs. Given one RGB frame from a head-worn camera, SpatialNet predicts:

(1) three presence flags (H, P, T): the wearer's hand is visible, the hand is pointing, and something is pointed at;
(2) the box $\mathbf{b}_h$ of the wearer's hand;
(3) the fingertip $\mathbf{p}$;
(4) the pointing direction $\mathbf{u}$ in the image plane;
(5) the box $\mathbf{b}_t$ of the pointed-at object, which we call the *target*.

Boxes are (c_x, c_y, w, h) and points (x, y), both normalised to $[0, 1]$. The flags combine into four scenes that an interface can act on: 000 *no hand*, 100 *hand, not pointing*, 110 *pointing, no target* and 111 *pointing at a target*.

What separates this from generic detection.

- *Ownership.* When several hands are visible, $\mathbf{b}_h$, $\mathbf{p}$ and $\mathbf{b}_t$ must refer to the wearer's hand, and a frame showing only other people's hands is *no hand*.
- *Budget.* The model must run on the glasses' own processor, within an interactive latency, without a second network or a server call.

Scope. We do not recognise *what* the target is. $\mathbf{b}_t$ is the region that a later step would take as input: an object recogniser, a visual question-answering model or an application. Localising the region first means that this step can run on a small crop instead of the full frame, and only when a pointing event occurs. The cost of that later step is outside the scope of this paper.

4 SpatialNet

Figure 2 shows the complete network. A shared trunk turns the frame into one feature map. Four heads then run in a fixed order (wearer's hand, fingertip, pointing direction, pointed target), and each head reads the shared map *at the location that the previous head predicted*. We call these heads *instance-anchored*: the fingertip is searched for around the hand that was found, the direction is computed from features of that hand and fingertip, and the target is searched for along the resulting ray. Three small gates decide which outputs are valid. Everything is a single forward pass; there is no external hand detector, no language input, and no hand box at test time.

4.1 Shared Trunk

The frame is resized to 160×160. A ShuffleNetV2 0.5× backbone [28] (stages 2–4 only; ImageNet-pretrained [8]) gives feature maps at strides 8, 16 and 32. Two BiFPN layers [36] merge them into one $192 \times 20 \times 20$ map at stride 8. Each fused map is smoothed by a depthwise 3×3 convolution, SiLU, and parameter-free SimAM attention [39]. We then append two coordinate channels (CoordConv [25]) and reduce to D=256 channels with a depthwise-separable convolution. Two ConvMixer-style residual blocks [37] follow, each made of a depthwise 5×5 and a pointwise convolution with residual connections. Their batch-norm scales start at zero, so each block is the identity at initialisation. The output is the shared map $S \in \mathbb{R}^{256 \times 20 \times 20}$, which all heads read.

A 20×20 grid gives 8-pixel cells at the model input. The heads recover points with sub-cell precision (Section 4.2), so a finer grid would mostly add compute. Sharing one map keeps the heads cheap: the largest head has 104K parameters (Table 2).

4.2 Locating One Instance: Peak-Windowed Soft-Argmax

Each head predicts a 20×20 logit map H and turns it into a point. A global soft-argmax [35] returns the probability-weighted mean over all cells. When two hands are visible, that mean falls *between* them. We therefore restrict the softmax to a $(2r{+}1) \times (2r{+}1)$ window $\mathcal{W}$ around the arg-max cell $k^\star$:

$$\pi_k = \frac{\mathbb{1}[k \in \mathcal{W}]\ \exp(\tau H_k)}{\sum_{j \in \mathcal{W}} \exp(\tau H_j)}, \qquad \hat{\mathbf{c}} = \sum_k \pi_k\, \mathbf{g}_k, \tag{1}$$

where $\mathbf{g}_k \in [0,1]^2$ is the normalised position of cell k and τ is a temperature. The estimate follows a single peak, is differentiable within the window, and is not limited to cell centres. We use $(\tau, r) = (8, 2)$ for the hand and the target and $(10, 1)$ for the fingertip. The peak logit $H_{k^\star}$ is also passed to the gates as a confidence cue.

4.3 Instance-Anchored Heads

Wearer's hand. A depthwise 3×3 and pointwise head gives hand-centre logits H^h, and Eq. (1) gives the centre $\mathbf{c}_h$. For the box size (w_h, h_h), we sample a 3×3 neighbourhood of S around $\mathbf{c}_h$

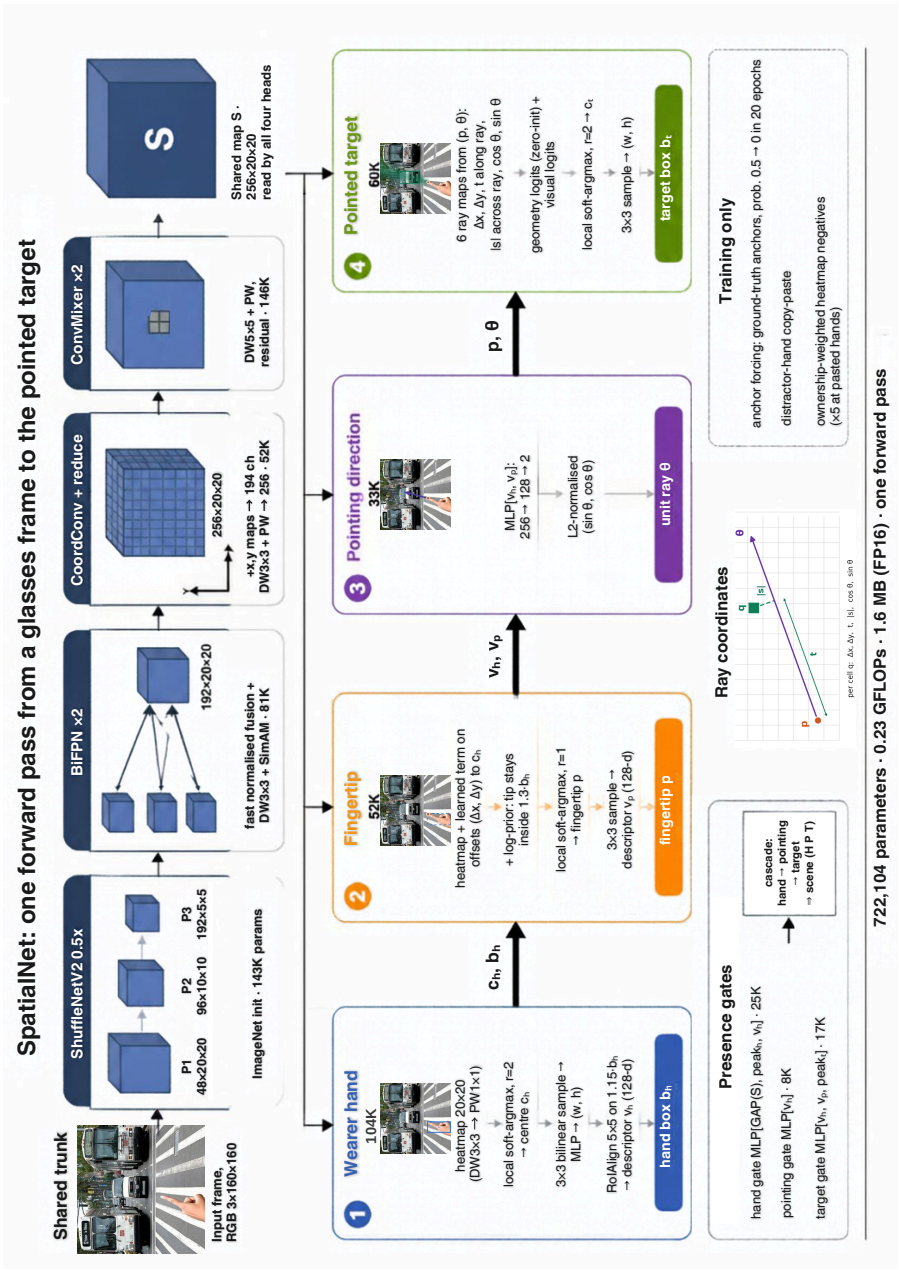


Fig. 2. SpatialNet. The trunk (top) builds one shared map S. Four heads (middle) run in order, each reading S where the previous one predicted: hand centre and box $\mathbf{c}_h, \mathbf{b}_h$; descriptors $\mathbf{v}_h, \mathbf{v}_p$; fingertip and direction $\mathbf{p}, \theta$. Bottom: presence gates, the ray coordinates of Eq. 4, and training-only mechanisms. Head panels give parameter counts; thumbnails show ground-truth labels of one test image.

bilinearly (one-cell spacing). A depthwise 3×3 convolution and a pointwise projection collapse it to 128 dimensions, and a two-layer MLP with a sigmoid maps it to the size, giving $\mathbf{b}_h = (\mathbf{c}_h, w_h, h_h)$. To describe *this* hand instance, we sample a 5×5 RoIAlign-style lattice [15] over $\mathbf{b}_h$ enlarged by 1.15. This lattice is collapsed to 128 dimensions and fused with the box coordinates into the hand descriptor $\mathbf{v}_h \in \mathbb{R}^{128}$.

Fingertip. The fingertip must belong to the hand just found. Its logits add three terms:

$$H^p = V^p(S) + G^p\big([S;\, \alpha_p(\mathbf{g} - \mathbf{c}_h)]\big) + B(\mathbf{b}_h), \tag{2}$$

- V^p is a visual head of the same form as the hand head.
- $\alpha_p(\mathbf{g} - \mathbf{c}_h)$ are two channels holding each cell's offset to the hand centre; α_p is a learnable scale.
- G^p is a small branch (a 1×1 convolution to 48 channels, then a 3×3 convolution to one channel) that learns where fingertips lie relative to the hand centre. Its last layer starts at zero.
- B is a fixed log-prior. It is zero inside the hand box dilated by δ=1.3 and falls linearly outside it:

$$B_k(\mathbf{b}_h) = -\frac{\lambda}{\tau_p}\left(\left[\frac{|g_{k,x} - c_{h,x}|}{\delta w_h/2} - 1\right]_+ + \left[\frac{|g_{k,y} - c_{h,y}|}{\delta h_h/2} - 1\right]_+\right), \tag{3}$$

  where $[\cdot]_+ = \max(\cdot, 0)$. With λ=4, the penalty after the softmax temperature τ_p is 4 logits for every half box-width outside the dilated box. No gradient flows from the prior into $\mathbf{b}_h$.

In the training labels, the fingertip lies inside the dilated box in 99.4% of pointing images, so the prior rarely contradicts a label, while it rules out the fingertips of other hands. Eq. (1) (with r=1) gives the fingertip $\mathbf{p}$, and a 3×3 sample around it gives the fingertip descriptor $\mathbf{v}_p \in \mathbb{R}^{128}$.

Pointing direction. An MLP ($256 \to 128 \to 2$, tanh) maps $[\mathbf{v}_h; \mathbf{v}_p]$ to a vector. We L2-normalise it into the unit direction $\mathbf{u} = (\cos\theta, \sin\theta)$ in image coordinates. The direction is supervised with the image-plane direction from the labelled fingertip to the labelled target centre (Section 4.5). $\mathbf{u}$ is therefore the direction *to the target* as it appears in the image, not the 3D orientation of the finger.

Pointed target: ray-conditioned search. This head carries the main idea of the model. From $\mathbf{p}$ and $\mathbf{u}$, every cell k receives six *ray coordinates* (inset in Figure 2):

$$\Delta_k = \alpha_t(\mathbf{g}_k - \mathbf{p}), \qquad t_k = \Delta_k \cdot \mathbf{u}, \qquad s_k = \big|\Delta_{k,x} \sin\theta - \Delta_{k,y} \cos\theta\big|, \qquad \cos\theta, \qquad \sin\theta. \tag{4}$$

These are:

- Δ_k (two channels): the scaled offset from the fingertip;
- t_k: the signed distance along the ray, negative behind the fingertip;
- s_k: the distance from the line of the ray;
- $\cos\theta$ and $\sin\theta$: the direction itself.

The target logits are

$$H^t = V^t(S) + G^t\big([S;\, R(\mathbf{p}, \theta)]\big), \tag{5}$$

where R stacks the six maps, and G^t (13K parameters) has the form of G^p and also starts at zero. The soft-argmax applies a softmax to τH^t, so $\pi_k \propto \exp(\tau V_k^t)\exp(\tau G_k^t)$: the geometry term acts as a learned *multiplicative prior* on the visual evidence. Because G^t starts at zero, training begins with a purely visual head, and the network uses the ray only as far as it lowers the loss.

We chose a learned prior over a hand-designed cone because the right shape depends on the situation. When the fingertip is on the object (about 70% of EgoPoint targets, Section 5), the target centre can lie in any direction from the fingertip, even slightly behind it. For distant targets, a

Table 2. Parameters per module. FLOPs are counted by PyTorch's `FlopCounterMode` (one multiply–accumulate = 2 FLOPs; bilinear sampling not counted) for one 160 × 160 frame. Sizes are serialised state dictionaries.

Module	Parameters	Role
ShuffleNetV2 0.5×, stages 2–4	143,136	backbone, ImageNet-pretrained
BiFPN ×2	81,426	multi-scale fusion to stride 8
CoordConv + reduce	52,310	194 → 256 channels
ConvMixer ×2	145,920	context mixing (5 × 5 depthwise)
Hand head	104,259	heatmap, size, descriptor $\mathbf{v}_h$
Fingertip head	51,826	incl. 12,913 in the hand-offset branch G^p
Direction MLP	33,154	$[\mathbf{v}_h; \mathbf{v}_p] \to \mathbf{u}$
Target head	60,404	incl. 13,105 in the ray branch G^t
Gates (hand / pointing / target)	24,769 / 8,321 / 16,577	presence logits
Learnable scales α_p, α_t	2	
Total	**722,104**	0.231 GFLOPs; 3.0 MB FP32; 1.6 MB FP16

narrow band along the ray is informative. The target centre $\mathbf{c}_t$ follows from Eq. (1). Its size comes from a 3×3 sample, as for the hand, giving $\mathbf{b}_t = (\mathbf{c}_t, w_t, h_t)$.

4.4 Presence Gates and Scene Decoding

Three MLP gates output logits for:

- *the wearer's hand is present*, from $[\mathrm{GAP}(S);\ H^h_{k^\star};\ \mathbf{v}_h]$;
- *the hand is pointing*, from $\mathbf{v}_h$ alone;
- *there is a pointed target*, from $[\mathbf{v}_h; \mathbf{v}_p; H^t_{k^\star}]$.

At inference, they are read as a cascade with threshold 0.5:

(1) If the hand gate is off, the frame is *no hand.*
(2) Otherwise, if the pointing gate is off, it is *hand, not pointing.*
(3) Otherwise, if the target gate is off, it is *pointing, no target.*
(4) Otherwise, it is *pointing at a target.*

The pointing and target gates read instance descriptors, not the whole image, so another person's pointing hand elsewhere in the frame cannot switch them on by itself. The hand gate also sees the global pool, so that a frame containing only other people's hands can be classed as having *no wearer's hand.* EgoPoint contains only the first and last of the four scenes (Section 5). We therefore evaluate the hand gate and make no claims about the pointing and target gates.

4.5 Training Objective

The loss is a weighted sum of 13 terms (weights in Appendix A). Each term is computed only on images whose label contains the parts it needs.

Supervised localisation.

- Hand box $\mathbf{b}_h$: CIoU loss [42].
- Fingertip $\mathbf{p}$: Wing loss [11], on normalised coordinates with w=0.5 and ϵ=0.1.
- Direction: $1 - \cos(\mathbf{u}, \mathbf{u}^\star)$, where $\mathbf{u}^\star$ is the unit vector from the labelled fingertip to the labelled target centre.
- Target box $\mathbf{b}_t$: SIoU loss [12] plus 1 − IoU.

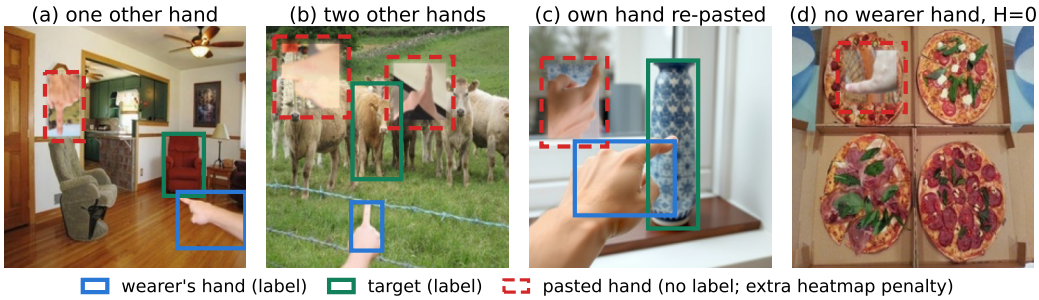


Fig. 3. Distractor-hand pasting, produced by the training code. Pasted hands (red, dashed) come from other images (a, b) or from the same image (c), are rotated so that they rarely enter from the bottom edge, and are never labelled. A no-hand image with a pasted hand keeps the label *no hand* (d).

Heatmaps. The penalty-reduced focal loss of CornerNet and CenterNet [18, 43] (α=2, β=4), on Gaussian targets at the hand centre, fingertip and target centre (σ = 1.5, 0.8 and 1.2 cells). These give each heatmap a direct signal, in addition to the gradient that reaches it through the soft-argmax.

Gates. Focal binary cross-entropy [24] (γ=2), each gate on its own population: the hand gate on all images, the pointing gate on hand images, and the target gate on pointing images. Positives are weighted 2× (3× for the hand gate), because a missed pointing gesture costs the user more than a false alarm that the interface can ignore.

Geometric consistency. Two terms use predictions only.

The *ray-alignment* term asks the predicted target centre to lie on the predicted ray, in front of the fingertip:

$$\mathcal{L}_{\text{ray}} = \left\| \mathbf{r} - (\mathbf{r} \cdot \hat{\mathbf{u}})\, \hat{\mathbf{u}} \right\| + 0.5 \left[-\mathbf{r} \cdot \hat{\mathbf{u}} \right]_+, \qquad \mathbf{r} = \hat{\mathbf{c}}_t - \hat{\mathbf{p}}. \tag{6}$$

- $\mathbf{u}^\star$ runs from the labelled fingertip to the labelled target centre, so the ground truth gives $\mathcal{L}_{\text{ray}} = 0$ exactly. The term therefore does not move the optimum; it only penalises predictions that disagree with each other.
- For a fixed angular error, the term grows with the fingertip–target distance, which puts more weight on far targets.

The *hand-axis* term is a hinge that keeps $\hat{\mathbf{u}}$ within 60° of the axis from the predicted hand centre to the predicted fingertip; it is zero inside this cone.

- It is only a weak plausibility prior. The labelled direction lies inside the cone in 68.6% of training images: 64.1% when the fingertip is on the object and 83.6% for far targets.
- For that reason it has the smallest weight of all 13 terms. Ablation A2 (Section 7.3) removes both consistency terms.

4.6 Learning Whose Hand It Is

No public pointing dataset labels other people's hands, and EgoPoint frames contain only the wearer's. We therefore create distractors during training (Figure 3). The approach follows copy-paste augmentation [9, 13], but with the opposite aim: pasted hands are *not* labelled, and the model is penalised for responding to them.

Pasting.

- *Frequency.* With probability 0.5 for images with a hand and 0.3 for images without one, we paste one or two hand crops taken from other training images.
- *Size.* Each crop is scaled to 0.45–1.0 of its source size.

- *Orientation.* With probability 0.7, the crop is rotated by 90, 180 or 270° and possibly mirrored. Unlike the wearer's arm, it therefore rarely enters from the bottom edge.
- *Placement.* The crop is centred in the upper 65% of the frame, with less than 15% overlap with the wearer's hand and the target.
- *Blending.* The crop is blended with a feathered alpha mask and a small colour jitter.
- *Self-paste.* In 30% of cases, the pasted crop is the wearer's own hand from the same image. Appearance alone then cannot tell the two hands apart, and the model has to use position, orientation and context.
- *Decoys.* With probability 0.15, we also paste an unlabelled patch that contains no hand, so that a pasted rectangle is not itself a cue.

Labels never change. In particular, a no-hand image with pasted hands remains *no hand*, which teaches the hand gate to answer for the wearer only.

Ownership-weighted negatives. In the hand-heatmap loss, the negative term near a pasted hand is multiplied by $1 + 4\,G_k$, where G_k is a Gaussian (σ=1.5 cells) at that hand's centre. A response on another person's hand thus costs up to five times as much as one on background.

The prior this encodes. The procedure assumes that, from a head-worn camera, the wearer's arm enters from the bottom of the frame, while other people's hands appear higher and in other orientations. This holds for most use of glasses but not always, for example when the wearer raises a hand above eye level or when a companion's hand is near the bottom of the view. Section 9 discusses this limitation.

4.7 Scheduled Anchor Forcing

Early in training, the predicted hand centre, fingertip and target centre are close to random, so the downstream heads would read features at meaningless locations. As in scheduled sampling for sequence models [3], we sometimes replace each anchor with its ground-truth location. This happens independently per image and per anchor, with a probability that decreases linearly from 0.5 to 0 over the first 20 epochs. From then on, training uses the model's own anchors, as at test time.

5 Data and Evaluation Protocol

5.1 EgoPoint

We train and test on the image set of EgoPoint-Ground [22], which we call EgoPoint. Its paper reports 15,338 images:

- 39% real captures with RayNeo V3 smart glasses (12 MP Sony IMX681 sensor);
- 39% synthetically composed;
- 22% generated by large multimodal models.

The release we use has 15,455 images and the split used by VistaRef [21] (3,072 test images). [Confirm with the EgoPoint-Ground authors that this release and split match the paper; the paper states a 7:2:1 train:val:test ratio, while the release is about 7:1:2. State the access terms and licence.] Each positive record shows one wearer's hand pointing at one target; negatives show no hand. Per-image source labels (real, composed or generated) are [not available in our copy; if obtained, also report results on the real subset].

Conversion. We convert each record to our label format (the three flags, hand box, fingertip and target box):

- Negatives become 000, and their object boxes are ignored.
- Positives with a hand box, an annotated fingertip and a target box become 111.

Table 3. EgoPoint after conversion. *On object*: the fingertip lies inside the target box; *near*: fingertip–box distance ≤ 0.08 of the image side; *far*: > 0.08.

				Pointing at target, by distance		
Split	Images	No hand (000)	Pointing at target (111)	On object	Near	Far
Train	10,669	1,535	9,134	6,511	1,995	628
Validation	1,524	211	1,313	922	297	94
Test	3,029	425	2,604	1,796	599	209
Total	15,222	2,171	13,051	9,229	2,891	931

- In three test records, the hand label was attached to an object and the hand was labelled as an object. We detected these by the fingertip position and swapped the labels back.
- We dropped 232 positives: 229 without an annotated fingertip (164 train, 22 validation, 43 test) and 3 without a hand box. All 229 are flagged as unclear in the source annotations. Spot checks showed open palms, held objects, or a hand box on the hand that was not pointing.

Table 3 summarises the result. Images are stored at 224×224; the model squashes every frame to its square 160×160 input, regardless of the original aspect ratio.

Consequences for evaluation.

- *Test split.* Our test split has 3,029 images, 43 positives fewer than the official one. Wherever we show numbers published on the official split, we say so; those numbers are not strictly comparable (Section 7.1).
- *Two scenes only.* EgoPoint contains only scenes 000 and 111. The pointing and target gates therefore never see a negative example, and we report neither their accuracy nor four-class scene accuracy.
- *Distance skew.* About 70% of targets are touched by the fingertip, and only 7% are far. Pooled results are dominated by touched targets, so we report every target metric by distance as well. The median distance from the fingertip to the target centre is 0.12, 0.17 and 0.32 of the image side for on-object, near and far targets (test split).

5.2 Metrics

- *Target and hand localisation.* We report mean IoU with a 95% bootstrap confidence interval (1,000 resamples [10]), and the share of images with IoU ≥ 0.3, 0.5 and 0.7. Target metrics are given overall and by distance.
- *Fingertip and direction.* Fingertip error is in pixels at the 160×160 input (1 px = 0.63% of the image side). Direction error is the angle between $\hat{\mathbf{u}}$ and $\mathbf{u}^\star$.
- *Hand presence.* For the hand gate, we report accuracy, missed hands, false alarms and AUROC, and the two-scene accuracy (no hand versus pointing at target), with Wilson intervals [38] for proportions.
- *No skipped images.* Localisation is scored on every image whose label contains the part, whatever scene the model predicted, so hard images cannot be skipped.

Ownership test. To test robustness to other people's hands, we paste one or two hands into every test image. The hands come from other test images, 30% are re-pasted copies of the wearer's own hand, and their positions are fixed by a seed. Labels stay unchanged. We report:

- *ownership accuracy*: the share of hand images in which the predicted hand centre is nearer the wearer's hand than any pasted hand;

- how often the predicted centre falls inside a pasted hand;
- the share of no-hand images that stay *no hand*.

These distractors follow the training distribution, so this test measures robustness to that synthetic distribution and may overstate robustness in real scenes. Section 7.4 therefore adds real frames.

Model selection. The reported checkpoint has the lowest validation loss. It was fixed before we looked at the test split, and the test split was evaluated once. [Confirm: loss weights and all hyper-parameters were set on the validation split only.]

6 Implementation and On-Glasses Deployment

6.1 Training

Optimisation.

- AdamW [27], weight decay 10^{-4}, batch size 32.
- One-cycle schedule [34]: peak learning rate 3×10^{-3} for the heads and 3×10^{-4} for the backbone, 10% warm-up, then cosine decay.
- Up to 180 epochs, stopping early after 25 epochs without a lower validation loss.
- Gradient clipping at 1.0, mixed precision, seed 42.

Training took xx.x h on one [GPU model] and stopped after [N] epochs.

Augmentation. Besides the distractor pasting of Section 4.6:

- horizontal flip (no vertical flip, because a wearer's hand entering from the top would contradict the ownership prior);
- shift, scale and rotation (±5%, ±10%, ±15°; probability 0.4);
- perspective warp (0.2);
- brightness and contrast (±20%; 0.5);
- Gaussian blur (0.2);
- mixup [41] between images of the same scene, with $\lambda \sim \mathrm{Beta}(0.2, 0.2)$ folded to ≥ 0.5 and the label of the dominant image.

All augmentation is on the fly; the dataset on disk is never modified. Appendix B lists all parameters.

Export. The model is exported to ONNX (opset 17, which bilinear grid sampling requires) and to FP16. The FP16 weights take 1.6 MB.

6.2 Deployment on Smart Glasses

We deploy SpatialNet on a smart-glasses developer unit [confirm what may be disclosed about this unit and anonymise for review; the spec sheet we have appears to be internal]. The unit has:

- a Snapdragon AR1 Gen 1 system-on-chip;
- a single 12 MP ultra-wide camera (105°) mounted in portrait orientation;
- a 160 mAh battery, in a frame weighing under 45 g;
- [operating system and version].

Pipeline. [Describe: runtime (e.g., ONNX Runtime / TFLite / QNN), execution unit (CPU, GPU or NPU) and precision; how frames are captured, resized to 160×160 and converted; when the model runs (every frame, or only after a trigger); and how the result reaches the user (audio, companion phone, or a downstream recogniser on the target crop).]

The camera delivers portrait frames, whereas most EgoPoint images are landscape. The model squashes both to a square, so the deployed input has a different aspect ratio from most training

images. The accuracy-parity check in Table 8 measures the combined effect of the on-device conversion; real-frame accuracy is assessed in Sections 7.4 and 8.

Measurement. [Fill in when T2 is done:]

- model latency: median and 90th percentile over at least 200 frames, after 20 warm-up frames;
- end-to-end latency, from camera frame to displayed or spoken result;
- sustained throughput over 60 s;
- peak memory of the application;
- battery drain over 10 minutes of continuous inference;
- accuracy parity: on 100 test images, the agreement of the deployed model's outputs with PyTorch.

7 Results

We answer five questions:

(1) How well does SpatialNet find the pointed-at object, compared with the published state of the art, vision–language models and simple references (Section 7.1)?
(2) Where does it fail, by distance and by head (Section 7.2)?
(3) Which design choices matter (Section 7.3)?
(4) Does it follow the wearer's hand when other hands are visible (Section 7.4)?
(5) How fast does it run on the glasses (Section 7.5)?

Unless stated otherwise, SpatialNet numbers are mean ± standard deviation over three training seeds on the 3,029 test images.

7.1 Pointed-Target Localisation

References. We compare SpatialNet with four kinds of reference.

- *State of the art, as published.* VistaRef [21] reports results on the same 3,072-image test split (ours omits 43 unclear images). EgoPoint-Ground's pointing-only (POG) results [22] come from a different setting: the methods receive the ground-truth hand box, and results are given for the real-image subset. We copy these numbers and do not re-run the models.
- *Vision–language models (VLMs).* We prompt [model names, versions and access dates; cite each] zero-shot on our 3,029 test images and ask for the box of the object that the camera wearer is pointing at (prompts and parsing in Appendix C).
  - An answer that cannot be parsed into a box counts as IoU 0.
  - Latency is the wall-clock time per image, measured [API or local GPU].
- *Simple references* computed from our predictions:
  - the mean training target box at a fixed position;
  - a box of mean training size centred on SpatialNet's predicted fingertip;
  - the same box centred on the ground-truth fingertip.

  These references test whether knowing where the fingertip is would be enough. It is not. Even at the true fingertip, a mean-size box reaches IoU≥0.5 in only 11.4% of test images, and 12.0% when it is shifted by the mean training offset. Target sizes vary widely (2.6% to 44% of the image area, 10th to 90th percentile), so the extent of the target has to be estimated as well.

Results. [Write two to four sentences after T1. Keep the comparison measured:]

- [Expect SpatialNet to be clearly below VistaRef in accuracy. Say so, then state the cost on each side: parameters, hardware and latency. VistaRef reports 22.9 images/s on a GPU; SpatialNet runs on the glasses in xx.x ms.]

Table 4. Target localisation on the EgoPoint test split: mean target IoU and the share of images (%) whose IoU reaches each threshold. Published works count IoU$>\tau$; we count IoU$\geq\tau$. *Far*: IoU ≥ 0.5 on the 209 far targets. Latency per image on the hardware named in the text; only SpatialNet runs on the glasses.
* BEiT-3-L backbone (226M parameters with BEiT-3-Base), setting "Train-Real", 3,072 test images.
† Given the ground-truth hand box; real-image subset of the test split.

Method	Params	IoU	≥0.3	≥0.5	≥0.7	Far	Latency
Published results (not re-run)							
VistaRef* [21], image + fixed prompt	>226M	82.0	97.0	93.9	82.6	–	–
SV-CoT† [22], image + hand box	>1B	–	69.4	63.9	55.6	–	–
Qwen3-VL-4B† [22], image + hand box	4B	–	67.1	61.2	53.2	–	–
Vision–language models: image + text prompt, zero-shot, our test split							
[VLM 1]	[size]	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x s
[VLM 2]	[size]	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x s
[VLM 3]	[size]	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x s
Simple references							
Mean training box, fixed position	0	27.4	41.3	9.6	0.9	4.3	–
Mean-size box at predicted fingertip	0.72M	xx.x	xx.x	xx.x	xx.x	xx.x	–
Mean-size box at true fingertip (oracle)	–	29.0	43.6	11.4	1.0	0.0	–
SpatialNet (image only)	0.72M	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x ms
± std over 3 seeds		xx.x	xx.x	xx.x	xx.x	xx.x	

Table 5. Accuracy of each head by fingertip–target distance (test split, pointing-at-target images). Fingertip error in pixels at the 160×160 input; direction error in degrees. Brackets: 95% bootstrap confidence intervals of the mean.

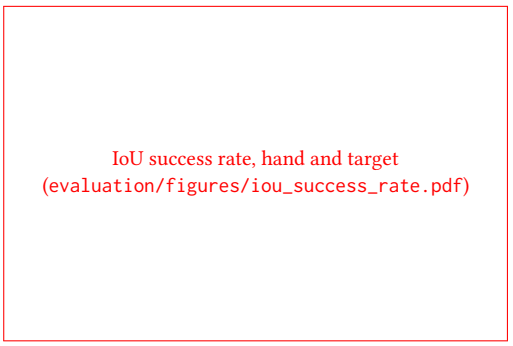
	Hand IoU		Fingertip (px)		Direction (°)		Target IoU	
Distance (images)	mean	≥0.5	mean	<5 px	mean	<15°	mean [CI]	≥0.5
On object (1,796)	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x [xx.x, xx.x]	xx.x
Near (599)	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x [xx.x, xx.x]	xx.x
Far (209)	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x [xx.x, xx.x]	xx.x
All (2,604)	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x [xx.x, xx.x]	xx.x

- [The VLMs are general, zero-shot models, while SpatialNet is trained on EgoPoint. The comparison shows what a specialised on-device model reaches against the general alternative a developer could call today; it does not show that VLMs are worse in general.]
- [Compare SpatialNet with the fingertip-centred references; the gap shows what the target head adds beyond finding the fingertip (extent on touched objects, the ray on near and far ones).]
- [Optional: a plot of accuracy (IoU≥0.5) against parameters on a log scale, with VistaRef, the POG results, the VLMs and SpatialNet.]

7.2 Per-Head Accuracy and Target Distance

Table 5 breaks the result down by head and by fingertip–target distance. Figure 4 shows the IoU success curves.

Hand presence. The hand gate separates no-hand from pointing frames with accuracy xx.x% [[CI]]. It misses xx.x% of hands, raises false alarms on xx.x% of no-hand frames, and has AUROC xx.x.



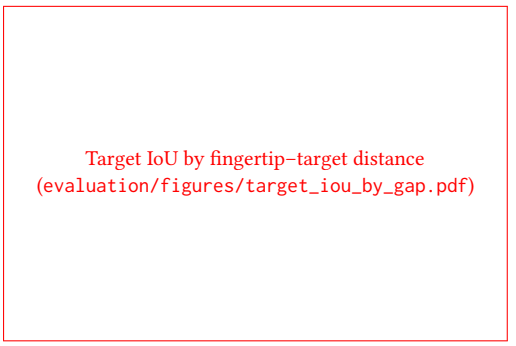


Fig. 4. Left: share of test images whose hand or target box reaches each IoU threshold. Right: mean target IoU and IoU ≥0.5 for touched, near and far targets. [Insert the figures produced by the evaluation script.]

Table 6. Ablations (one seed each). Differences from the full model with 95% paired-bootstrap intervals; intervals that exclude 0 are in bold. Ownership: ownership accuracy on the synthetic multi-hand test.

Variant	Target IoU	≥0.5	Far ≥0.5	Hand IoU	Fingertip (px)	Ownership (%)
Full model (seed 42)	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x
A1 no distractors / ownership weight	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x
A2 no geometric consistency terms	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x
A3 no ray conditioning	xx.x	xx.x	xx.x	xx.x	xx.x	xx.x

Results. [Two to four sentences. Expected pattern to verify: the target IoU drops from on-object to far targets, and the direction error matters most for far targets. Name the head that limits far targets (fingertip, direction or target box), using the table.]

7.3 Ablations

We retrain the model without one component at a time, with the seed and settings of the main run:

- **A1** no distractor pasting and no ownership weighting;
- **A2** no geometric consistency terms ($\mathcal{L}_{\text{ray}}$ and the hand-axis term);
- **A3** no ray conditioning of the target head (no G^t; 13,105 fewer parameters).

[Optional A4: no instance-anchored chain (heads read the globally pooled vector).] Each ablation was trained once. Instead of seed variation, we report a paired bootstrap over test images: we resample the 2,604 images and recompute the difference between the full model and the ablation on the same images, which yields a 95% confidence interval for each difference.

Results. [Write one claim per row, only as strong as its interval:]

- [A3 tests contribution C1. If far-target IoU does not drop significantly without G^t, say so and weaken C1 in the introduction and abstract.]
- [A2 tests whether the two consistency terms help, mostly on far targets.]
- [A1 should mainly affect ownership; report any cost on clean-image accuracy.]

The loss weights were not searched (Appendix A): A2 covers the two non-standard terms, and the remaining eleven are standard box, keypoint, heatmap and focal losses.

Table 7. Ownership on test images with one or two pasted hands (synthetic), and on real frames with other people's hands (T5). All values in %. *On other hand*: predicted hand centre inside a pasted hand. *No hand kept*: no-hand images with pasted hands still classed as no hand.

	Ownership accuracy	On other hand	No hand kept	Target IoU ≥0.5
Full model, clean test images	–	–	–	xx.x
Full model, pasted hands	xx.x	xx.x	xx.x	xx.x
A1, pasted hands	xx.x	xx.x	xx.x	xx.x
Full model, real frames ([N])	xx.x [xx.x, xx.x]	xx.x	–	–

Six real frames from the glasses with another person's hand in view: predicted wearer's hand, fingertip, ray and target (T5). Include at least one failure.

Fig. 5. Real frames with other people's hands. [Insert after T5.]

7.4 Following the Wearer's Hand

Synthetic distractors. Table 7 reports the multi-hand test of Section 5.2 for the full model and A1. Because the distractors follow the training distribution, these numbers show that the model has learned what it was trained on. They do not show robustness to real bystanders.

Real frames. [Fill in when T5 is done:]

- We recorded [N] clips with the glasses in which another person's hand is visible, sometimes pointing, with [consent procedure], and sampled [100–150] frames.
- Two annotators independently labelled the predicted hand in each frame as the wearer's, another person's, or missed.
- We report ownership accuracy with a Wilson interval and inter-annotator agreement (Cohen's κ).

Figure 5 shows examples.

7.5 On-Device Performance

Table 8 reports cost on a desktop CPU and on the glasses.

Results. [Two or three sentences: model latency against the frame interval of the camera, end-to-end latency against a 100–200 ms interactive budget, and the gap to the VLM latencies of Table 4. If the end-to-end time is dominated by capture or conversion rather than by the model, say so.]

7.6 Qualitative Results and Failure Analysis

Figure 6 shows typical successes and failures. We sorted the [N] test images with target IoU < 0.3 into failure types by inspection:

- [wrong object along the correct ray (similar objects in a row, as in Figure 1c);]

Table 8. Inference cost at batch size 1 and 160 × 160 input. Desktop numbers use PyTorch eager mode and are given for reference only. [CPU model].

Platform and measure	Runtime / precision	Median	p90
Desktop CPU, 1 thread	PyTorch FP32	xx.x ms	xx.x ms
Desktop CPU, all threads	PyTorch FP32	xx.x ms	xx.x ms
Glasses, model only	[runtime, unit, precision]	xx.x ms	xx.x ms
Glasses, camera frame to result	[same]	xx.x ms	xx.x ms
Glasses, sustained throughput (60 s)			xx.x frames/s
Glasses, peak memory / battery per 10 min			xx.x MB / xx.x%
Glasses vs. PyTorch, 100 test images		xx.x% same scene; box IoU xx.x	

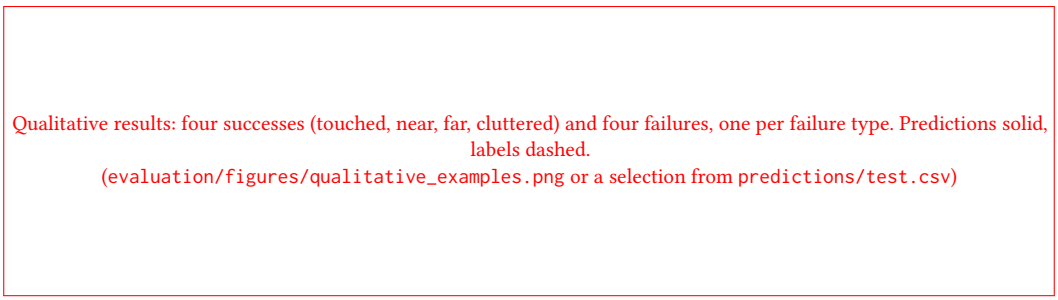


Fig. 6. Successes (top) and failures (bottom) on EgoPoint test images. [Insert after the main run.]

- [correct object but box too small or too large (object extent, e.g., a whole table versus the cup on it);]
- [wrong direction (fingertip or hand partly out of view);]
- [missed or wrong hand;]
- [questionable label.]

[Give a count for each type.]

8 User Study

[If T6 is not run, delete this section, move a short "deployment observations" paragraph to Section 9, and list the user study as future work. Every number and procedural detail below must come from the actual study.]

Goal and design. We tested whether pointing with SpatialNet works for real people on the glasses, and how quickly.

- *Participants.* [12] participants ([age range, gender, handedness, prior AR or smart-glasses use]) took part under [ethics approval or exemption, consent procedure].
- *Setup.* [10–15] everyday objects placed near (<1 m) and far (2–4 m) from the participant.
- *Task.* In each trial, the experimenter named an object, the participant pointed at it, and the glasses [highlighted or spoke] the localised target.
- *Trials.* [20] trials per participant (half near, half far), in [counterbalanced] order. [Optional second condition: the same task with a cloud VLM.]
- *Logging.* Each trial logged the named object, the localised object, correctness and the time from the start of the gesture to the result.

- *Questionnaires.* After the trials, participants completed the System Usability Scale [5] and the raw NASA-TLX [14], followed by a short interview.

Results.

- *Task success.* xx.x% [[Wilson CI]] overall: xx.x% for near objects and xx.x% for far ones.
- *Response time.* The median time from gesture to result was xx.x s.
- *Usability.* The SUS score was xx.x (benchmark 68), and the raw TLX xx.x.
- *Comparison.* [With two conditions: a paired Wilcoxon signed-rank test on success and time.]
- *Interviews.* [Three or four themes with short quotes, e.g., when pointing felt natural, failures participants noticed, privacy around others.]

9 Discussion

What the design contributes. [Rewrite once Table 6 is filled in, keeping only the statements its intervals support.] The architecture rests on three choices:

- *Anchoring.* Every head reads features where the previous head looked, so the chain stays on one hand instance.
- *Learned, safe geometry.* The ray enters as a zero-initialised log-prior. It cannot hurt at the start of training and is used only as far as it helps.
- *Ownership from training data.* Wearer-awareness comes from distractor pasting, not from extra labels.

These choices add little cost: the ray branch has 13K parameters, and the hand-offset branch 13K.

Why most targets are touched, and what that implies. Where people point at distant objects depends on the eye as well as the arm: modelling the dominant eye makes pointing rays markedly more accurate [32], and common ray-casting models show systematic offsets [29]. If wearers align the fingertip with the target from the eye, a camera next to the eye sees the target at or just behind the fingertip. This is consistent with about 70% of EgoPoint targets containing the fingertip in the image, although part of EgoPoint is staged or generated [confirm]. It has two consequences:

- For such targets, the fingertip tells *where* the object is but not *how large* it is: a mean-size box at the true fingertip reaches IoU≥0.5 in only 11.4% of test images (Table 4). On touched objects, the target head mainly has to estimate the object's extent.
- The ray matters for the remaining near and far targets, where the finger points away from the line of sight. This subset is small (7% far), so it needs its own metrics rather than pooled averages.

Limitations.

- *Two scenes.* The pointing and target gates were never trained or tested against negatives (Section 5). In deployment, a non-pointing hand may therefore be read as pointing, and the cascade's middle states are untested. A dataset with non-pointing hands and pointing without a target is needed before relying on them.
- *Ownership.* Our distractors are synthetic and encode the prior that the wearer's arm enters from the bottom of the view. The real-frame check (Section 7.4) is small. A wearer pointing upward from a raised arm, or a companion's hand close to the wearer's, can break the prior.
- *A 2D ray.* The ray lives in the image plane. Objects at different depths along the same image ray are told apart only by appearance, and the eye–camera offset is ignored. Depth, or a second view, could resolve such cases.
- *Single frames.* The model has no temporal context. Detecting a pointing *event*, as opposed to a pointing frame, and smoothing the target over a short window are left to the application.

- *Dataset coverage.* Results come from one dataset with mostly landscape, partly synthetic images and unknown demographic coverage (for example, skin tone and handedness). Flip augmentation covers left hands only geometrically. Evaluation on real, diverse glasses footage is the most important next step.
- *Baselines.* We copied the published state-of-the-art results instead of re-running those models, and two of them use different inputs or subsets. The VLMs were used zero-shot with [one prompt]. Large grounding models are clearly more accurate. Our claim is the operating point, with accuracy reported next to its cost, not superiority.

Implications for glasses interfaces. A model of this size can run continuously and act as a gate for more expensive components:

- *Crops instead of frames.* Only when the glasses detect a pointing frame with a confident target does a recogniser or assistant run, and then on the target crop instead of the full frame. This reduces computation, bandwidth and the amount of bystander imagery that leaves the device. [Quantify if T2 measured it; otherwise keep as a design implication.]
- *Confirmation for far targets.* Because far targets are where errors concentrate (Section 7.2), interfaces should confirm the selection, for example by naming the object or highlighting it, before acting on it.

10 Ethics and Privacy

Bystanders. A head-worn camera sees people who did not consent to being recorded.

- SpatialNet runs on the glasses, so localisation needs no frame to leave the device.
- A downstream step can be limited to the target crop and to moments when the wearer points.
- SpatialNet is trained to ignore other people's hands. It does not detect, identify or track people.

On-device processing still does not remove the need for visible recording indicators; the developer unit has user- and bystander-facing LEDs [confirm whether this may be stated].

Data. EgoPoint is used under its licence [name the licence; state whether the converted labels may be shared]. The images shown in this paper come from its test and training splits. [T5 and T6: describe consent, the ethics approval or exemption (with reference number after review), data retention, and the avoidance of bystanders' faces.]

Availability. [Anonymised link to the code, conversion script, trained weights and evaluation outputs.]

11 Conclusion

SpatialNet localises the object a smart-glasses wearer points at, from a single frame, with 0.72M parameters and one forward pass. It does so by chaining instance-anchored heads (hand, fingertip, direction, target) and by conditioning the target search on ray coordinates derived from its own predictions. Training with pasted, unlabelled hands teaches it to follow the wearer's hand. [One sentence with the key T1 and T2 numbers.] The main open problems are pointing scenes that EgoPoint does not contain, real bystander hands, and far targets along ambiguous rays. We expect progress on them to come from richer egocentric pointing data at least as much as from larger models.

References

[1] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. 2025. Qwen2.5-VL Technical Report. arXiv preprint arXiv:2502.13923.

[2] Sven Bambach, Stefan Lee, David J. Crandall, and Chen Yu. 2015. Lending A Hand: Detecting Hands and Recognizing Activities in Complex Egocentric Interactions. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 1949–1957.

[3] Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. 2015. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 28. 1171–1179.

[4] Richard A. Bolt. 1980. "Put-That-There": Voice and Gesture at the Graphics Interface. In *Proceedings of the 7th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '80)*. ACM, 262–270.

[5] John Brooke. 1996. SUS: A "Quick and Dirty" Usability Scale. In *Usability Evaluation in Industry*, Patrick W. Jordan, B. Thomas, Bernard A. Weerdmeester, and Ian L. McClelland (Eds.). Taylor & Francis, 189–194.

[6] Yixin Chen, Qing Li, Deqian Kong, Yik Lun Kei, Song-Chun Zhu, Tao Gao, Yixin Zhu, and Siyuan Huang. 2021. YouRefIt: Embodied Reference Understanding with Language and Gesture. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.

[7] Yura Choi, Roy Miles, Rolandos Alexandros Potamias, Ismail Elezi, Jiankang Deng, and Stefanos Zafeiriou. 2026. Do You See What I Am Pointing At? Gesture-Based Egocentric Video Question Answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

[8] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. ImageNet: A Large-Scale Hierarchical Image Database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 248–255.

[9] Debidatta Dwibedi, Ishan Misra, and Martial Hebert. 2017. Cut, Paste and Learn: Surprisingly Easy Synthesis for Instance Detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 1301–1310.

[10] Bradley Efron and Robert J. Tibshirani. 1994. *An Introduction to the Bootstrap*. Chapman and Hall/CRC.

[11] Zhen-Hua Feng, Josef Kittler, Muhammad Awais, Patrik Huber, and Xiao-Jun Wu. 2018. Wing Loss for Robust Facial Landmark Localisation with Convolutional Neural Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2235–2245.

[12] Zhora Gevorgyan. 2022. SIoU Loss: More Powerful Learning for Bounding Box Regression. arXiv preprint arXiv:2205.12740.

[13] Golnaz Ghiasi, Yin Cui, Aravind Srinivas, Rui Qian, Tsung-Yi Lin, Ekin D. Cubuk, Quoc V. Le, and Barret Zoph. 2021. Simple Copy-Paste Is a Strong Data Augmentation Method for Instance Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2918–2928.

[14] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Human Mental Workload*, Peter A. Hancock and Najmedin Meshkati (Eds.). Advances in Psychology, Vol. 52. North-Holland, 139–183.

[15] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. 2017. Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 2961–2969.

[16] Yichao Huang, Xiaorui Liu, Xin Zhang, and Lianwen Jin. 2016. A Pointing Gesture Based Egocentric Interaction System: Dataset, Approach and Application. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*.

[17] Varun Jain, Gaurav Garg, Ramakrishna Perla, and Ramya Hebbalaguppe. 2019. GestARLite: An On-Device Pointing Finger Based Gestural Interface for Smartphones and Video See-Through Head-Mounts. AAAI 2019 Workshop on Plan, Activity, and Intent Recognition. arXiv:1904.09843.

[18] Hei Law and Jia Deng. 2018. CornerNet: Detecting Objects as Paired Keypoints. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 734–750.

[19] Jaewook Lee, Jun Wang, Elizabeth Brown, Liam Chu, Sebastian S. Rodriguez, and Jon E. Froehlich. 2024. GazePointAR: A Context-Aware Multimodal Voice Assistant for Pronoun Disambiguation in Wearable Augmented Reality. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, Article 408. doi:10.1145/3613904.3642230

[20] Chentao Li, Zirui Gao, Mingze Gao, Yinglian Ren, Jianjiang Feng, and Jie Zhou. 2026. Do MLLMs Understand Pointing? Benchmarking and Enhancing Referential Reasoning in Egocentric Vision. arXiv preprint arXiv:2604.21461.

[21] Ling Li, Zhizhen Cai, Xinkun Wu, Ziyu Zhu, Jiaqing Lyu, Bowen Liu, and Zhidong Deng. 2026. VistaRef: Boosting Visual Spatial Orientation Awareness for Pointing-to-Object Detection. arXiv preprint arXiv:2606.24498.

[22] Ling Li, Bowen Liu, Zinuo Zhan, Peng Jie, Jianhui Zhong, Kenglun Chang, and Zhidong Deng. 2026. Beyond Language: Grounding Referring Expressions with Hand Pointing in Egocentric Vision. arXiv preprint arXiv:2603.26646.

[23] Ling Li, Bowen Liu, Zinuo Zhan, Jianhui Zhong, Ziyu Zhu, Bingcai Wei, Kenglun Chang, and Zhidong Deng. 2026. PointVG-R: Internalizing Geometric Reasoning in MLLMs for Precise Pointing Localization via Visual Chain of Thought. arXiv preprint arXiv:2606.24539.

[24] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal Loss for Dense Object Detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 2980–2988.

[25] Rosanne Liu, Joel Lehman, Piero Molino, Felipe Petroski Such, Eric Frank, Alex Sergeev, and Jason Yosinski. 2018. An Intriguing Failing of Convolutional Neural Networks and the CoordConv Solution. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31.

[26] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. 2024. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. In *Proceedings of the European Conference on Computer Vision (ECCV)*.

[27] Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations (ICLR)*.

[28] Ningning Ma, Xiangyu Zhang, Hai-Tao Zheng, and Jian Sun. 2018. ShuffleNet V2: Practical Guidelines for Efficient CNN Architecture Design. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 116–131.

[29] Sven Mayer, Katrin Wolf, Stefan Schneegass, and Niels Henze. 2015. Modeling Distant Pointing for Compensating Systematic Displacements. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15)*. ACM, 4165–4168. doi:10.1145/2702123.2702332

[30] Shu Nakamura, Yasutomo Kawanishi, Shohei Nobuhara, and Ko Nishino. 2023. DeePoint: Visual Pointing Recognition and Direction Estimation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 20577–20587.

[31] Zhiliang Peng, Wenhui Wang, Li Dong, Yaru Hao, Shaohan Huang, Shuming Ma, and Furu Wei. 2023. Kosmos-2: Grounding Multimodal Large Language Models to the World. arXiv preprint arXiv:2306.14824.

[32] Katrin Plaumann, Matthias Weing, Christian Winkler, Michael Müller, and Enrico Rukzio. 2018. Towards Accurate Cursorless Pointing: The Effects of Ocular Dominance and Handedness. *Personal and Ubiquitous Computing* 22 (2018). doi:10.1007/s00779-017-1100-7

[33] Dandan Shan, Jiaqi Geng, Michelle Shu, and David F. Fouhey. 2020. Understanding Human Hands in Contact at Internet Scale. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9869–9878.

[34] Leslie N. Smith and Nicholay Topin. 2019. Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates. In *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications (Proceedings of SPIE, Vol. 11006)*.

[35] Xiao Sun, Bin Xiao, Fangyin Wei, Shuang Liang, and Yichen Wei. 2018. Integral Human Pose Regression. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 529–545.

[36] Mingxing Tan, Ruoming Pang, and Quoc V. Le. 2020. EfficientDet: Scalable and Efficient Object Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10781–10790.

[37] Asher Trockman and J. Zico Kolter. 2023. Patches Are All You Need? *Transactions on Machine Learning Research* (2023).

[38] Edwin B. Wilson. 1927. Probable Inference, the Law of Succession, and Statistical Inference. *J. Amer. Statist. Assoc.* 22, 158 (1927), 209–212.

[39] Lingxiao Yang, Ru-Yuan Zhang, Lida Li, and Xiaohua Xie. 2021. SimAM: A Simple, Parameter-Free Attention Module for Convolutional Neural Networks. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research, Vol. 139)*. 11863–11874.

[40] Haoxuan You, Haotian Zhang, Zhe Gan, Xianzhi Du, Bowen Zhang, Zirui Wang, Liangliang Cao, Shih-Fu Chang, and Yinfei Yang. 2024. Ferret: Refer and Ground Anything Anywhere at Any Granularity. In *International Conference on Learning Representations (ICLR)*.

[41] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. 2018. mixup: Beyond Empirical Risk Minimization. In *International Conference on Learning Representations (ICLR)*.

[42] Zhaohui Zheng, Ping Wang, Wei Liu, Jinze Li, Rongguang Ye, and Dongwei Ren. 2020. Distance-IoU Loss: Faster and Better Learning for Bounding Box Regression. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34. 12993–13000.

[43] Xingyi Zhou, Dequan Wang, and Philipp Krähenbühl. 2019. Objects as Points. arXiv preprint arXiv:1904.07850.

A Loss Weights

Table 9 lists the 13 loss terms and their weights. [One true sentence on how the weights were set, e.g.: set by hand during development so that the weighted terms had similar magnitudes early in training, judged on the validation split only; no search on the test split.]

Table 9. Loss terms, their supervision, and weights.

Term	Loss	Supervision	Weight
Hand box	$1 - \mathrm{CIoU}$	hand box	0.10
Fingertip	Wing (w=0.5, ϵ=0.1)	fingertip	0.18
Direction	$1 - \cos$ine	fingertip $\rightarrow$ target centre	0.03
Target box	$1 - \mathrm{SIoU}$	target box	0.45
Target box, auxiliary	$1 - \mathrm{IoU}$	target box	0.10
Hand gate	focal BCE, positives $\times 3$	H	0.08
Pointing gate	focal BCE, positives $\times 2$	P (hand images)	0.10
Target gate	focal BCE, positives $\times 2$	T (pointing images)	0.06
Ray alignment	distance to ray + $0.5 \cdot$ behind	none (predictions only)	0.08
Hand axis	hinge outside a 60° cone	none (predictions only)	0.02
Hand heatmap	penalty-reduced focal, ownership-weighted	hand centre	0.10
Fingertip heatmap	penalty-reduced focal	fingertip	0.10
Target heatmap	penalty-reduced focal	target centre	0.10

Table 10. Training-time augmentation, applied on the fly; each training image is seen once per epoch as a new random variant. Validation and test images are only resized and normalised.

Step	Setting
Distractor hands	1–2 hands with probability 0.5 (hand images) or 0.3 (no-hand images); scale 0.45–1.0; rotation by 90/180/270° and optional mirror with probability 0.7; centre in the top 65% of the frame; overlap with the wearer's hand and the target below 15%; feathered alpha (12% of the crop size); colour jitter; 30% self-paste; pool of 512 donor crops, half from pointing hands
Decoy patch	unlabelled non-hand patch from a no-hand image, with probability 0.15
Horizontal flip	0.5
Shift / scale / rotate	±5% / ±10% / ±15°, probability 0.4
Perspective	scale 0.02–0.05, probability 0.2
Brightness / contrast	±20%, probability 0.5
Gaussian blur	kernel 3–5, probability 0.2
Mixup	within the same scene, $\lambda \sim \mathrm{Beta}(0.2, 0.2)$ folded to ≥ 0.5, label of the dominant image, every batch
Anchor forcing	ground-truth anchors with probability 0.5 decaying to 0 over 20 epochs

B Augmentation Parameters

C Vision–Language Model Prompts

[For each model: name, version or snapshot, access date, and decoding settings (temperature, maximum tokens); the exact prompt; the expected output format and how it was parsed into a box (coordinate convention, normalisation); the share of answers that could not be parsed; and the latency measurement setup.]